

Four-Session Workshop Plan for Evaluating AI and New Data Sources in Consumer Lending

Executive summary

This four-session (8 total hours), time-boxed workshop is designed for a 5-person cross-functional team to evaluate where **AI** (including advanced analytics and carefully governed generative AI) and **new data sources** can improve a consumer lending business **without compromising fair lending, privacy, explainability, or model risk management**. The plan is structured around the supervisory expectations embedded in interagency model risk management guidance (commonly referenced as SR 11-7), which emphasizes **sound model development/implementation/use, effective independent validation, and strong governance**. ¹

Over four 2-hour sessions you will produce: a prioritized AI/data use-case backlog; a “permissible data” shortlist with explicit exclusions; a feature inventory and risk assessment matrix; a minimal technical experiment plan (baseline + incremental alternative data tests) with metrics (including calibration, stability/drift, and fairness testing); governance checkpoints and documentation requirements; and a pilot-ready handoff package for validation and compliance review. The plan is a **project plan**, not legal advice.

If your broader program requires an “evidence-driven” presentation deliverable (e.g., for an executive academy or capstone), the outputs in Sessions 3–4 can be organized into a concise deck plus “behind-the-scenes” documentation of AI tool usage and human edits. `filecite\turn0file0`

Design principles and guardrails

Regulatory and governance anchors used to shape the workshop

This workshop plan intentionally “bakes in” requirements from primary supervisory and regulatory sources so that the team does not treat compliance as an afterthought:

- **Model risk management:** Interagency supervisory guidance (SR 11-7) lays out expectations for development/implementation, validation, and governance, and is reinforced by the **OCC Model Risk Management** handbook materials used in examinations. ²
- **Adverse action explainability:** Regulation B requires adverse action notices with **specific, principal reasons**; generic statements like “failed to achieve a qualifying score” are insufficient. ³
The CFPB has explicitly stated there is **no “special exemption”** for AI: creditors must provide accurate and specific reasons even when using complex models and large datasets (including data harvested from “consumer surveillance”). ⁴
- **Alternative data:** The interagency statement on alternative data highlights that firms are using or considering alternative data not only for underwriting but also for fraud detection, marketing,

pricing, servicing, and account management—while emphasizing consumer protection risks that must be managed. ⁵

- **Third-party / vendor risk:** Interagency third-party risk management guidance establishes a lifecycle approach (planning, due diligence, contracting, monitoring, and termination) that is directly relevant when using data aggregators, income verification services, model platforms, or AI vendors. ⁶
- **Privacy and security:** Regulation P (GLBA privacy) governs treatment and disclosures of nonpublic personal information, and the FTC Safeguards Rule provides concrete expectations for information security programs and safeguards for customer information. ⁷
- **AI risk framing** (optional but useful): NIST AI RMF 1.0 offers a structured vocabulary and lifecycle approach (“Map, Measure, Manage, Govern”) for identifying and managing AI risks in context. ⁸

Non-negotiable guardrails for “new data sources” and AI use

1) Hard exclusions for underwriting drivers

Treat the following as presumptively prohibited/high-risk for creditworthiness decisions due to fairness, privacy, and explainability concerns—especially when they could act as proxies for protected classes or are difficult to translate into specific adverse action reasons:

- Social graph / friendship network / “people you interact with,” influence scores, or social engagement signals
- Family relationship data beyond what is necessary for a product structure (e.g., joint applicants)
- Inheritance potential or “expected wealth transfers”
- Highly granular location trails, “consumer surveillance” signals, or psychographic/behavioral inferences unrelated to repayment capacity

The CFPB has highlighted that data harvested from consumer surveillance may be used in underwriting and that consumers can be denied for reasons they may not view as relevant; it simultaneously emphasized the need for specific and accurate reasons for adverse action with no AI exemption. ⁹

2) Explainability must be designed, not retrofitted

Regulation B commentary notes that more than four reasons is generally not helpful, and reasons must accurately describe factors actually considered or scored. ¹⁰

Therefore, reason-code feasibility is a *Session 1–3 design constraint*, not a “Session 4 documentation step.”

3) Separate fraud/ID from creditworthiness unless explicitly justified

Device and identity signals tend to be explainability-poor and proxy-prone if used as creditworthiness drivers. A defensible approach is a **fraud gate** (approve/decline for fraud risk or step-up verification) upstream of a **credit risk model** for repayment risk, with governance around each. This also helps keep adverse action reasons accurate and legally aligned. ¹¹

4) Cash-flow data is the “most research-backed” alternative-data candidate—but still requires your own testing

FinRegLab’s empirical work (supported by analyses conducted with Charles River Associates ¹²) found cash-flow measures predictive in the studied samples and did not observe disparate impact in those specific settings; however, that is not a universal safe harbor and does not replace institution-specific fair lending and governance analysis. ¹³

FinRegLab’s 2025 research further examines machine learning plus cash-flow data and discusses

explainability and fairness considerations, reinforcing the need to treat this as an evidence-driven experimentation program rather than an assumption. ¹⁴

Team roles, responsibilities, and required pre-work

This section defines how a 5-person team can cover the full lifecycle: business framing → data feasibility → minimal modeling tests → governance and pilot readiness.

Roles and responsibilities for the 5-person team

- **Workshop Lead (you: Senior Credit Risk / Model Governance Lead)**
Owns scope, facilitation, decisions, and the governance spine (model risk tiering, sign-off gates, decision log). Aligns outputs to SR 11-7 style expectations and OCC MRM handbook themes. ¹
- **Data Engineer / Data Architect**
Owns data inventory, feasibility, integration approach, lineage, quality checks, and secure environment constraints (including what can/cannot be shared with AI tools).
- **Modeler (Credit Risk Data Scientist / Quant)**
Owns minimal model design, label definitions, leakage checks, baseline metrics, calibration, stability/drift tests, and fairness measurement plan.
- **Compliance & Legal Liaison (Fair Lending + Privacy + Third-Party Governance)**
Owns guardrails, documentation requirements, vendor risk checkpoints, and adverse action/readability constraints; flags where counsel review is required (without providing legal advice).
- **Product & Operations Lead (Lending Ops / Servicing / Fraud Ops / CX)**
Owns process mapping, operational impact sizing, implementation constraints, and how decisions translate into workflows (verification steps, adverse action notices, exception handling, servicing/collections playbooks).

Pre-work checklist by role (what each person must gather)

Pre-work is critical because the sessions are short. Each person should spend ~2–5 hours before Session 1 collecting inputs; the data engineer and modeler may need more if extracting baseline metrics.

Role	Required pre-work inputs to bring	Why it matters	Suggested effort
Workshop Lead	Current lending strategy and risk appetite; target product(s); key metrics (approval rate, loss rate, ROA/ROE proxy); current model inventory entry (if exists) and governance artifacts	Anchors workshop decisions to business objectives and existing governance expectations (SR 11-7 pillars) ¹	3–5 hrs

Role	Required pre-work inputs to bring	Why it matters	Suggested effort
Data Engineer / Architect	Data inventory: bureau/CRA feeds, application data, account/servicing data, bank transaction access (if any), income/employment verification, alternative repayment data (rent/utilities), fraud/ID datasets; data dictionaries; join keys; latency; known quality issues	Enables realistic, auditable “data architecture” decisions; prevents selecting data that can’t be operationalized or explained	4–8 hrs
Modeler	Baseline model performance summary (AUC/KS, calibration, approval/loss by score band); definition of default and observation windows; sample bias notes; existing reject inference approach (if any); prior fairness testing outputs (if available)	Establishes “baseline” so alternative data/AI value is incremental and measurable	4–8 hrs
Compliance & Legal Liaison	Current adverse action templates and reason-code library; privacy notices/data sharing constraints (Reg P); vendor oversight process; list of known prohibited/high-risk data categories; summary of key requirements: Regulation B specific reasons; CFPB circulars on complex models and sample forms ¹⁵	Sets non-negotiable constraints: specificity, fairness posture, privacy, third-party risk	3–6 hrs
Product & Ops	End-to-end process map (origination → verification → funding → servicing → collections); cycle-time bottlenecks; manual decision points; fraud losses and escalation rates; customer pain points and complaint drivers	Ensures AI evaluation includes operating model impact, not just statistical lift	3–5 hrs

Shared pre-read pack (30–60 minutes each)

Assign these to all five members to align language and expectations:

- SR 11-7 supervisory guidance overview and validation/governance expectations ¹⁶
- CFPB circular on complex algorithms and adverse action specificity (2022-03) ¹⁷
- CFPB circular on proper use of sample adverse action forms (2023-03) ¹⁸
- Interagency statement on alternative data ¹⁹
- Interagency third-party risk guidance ²⁰
- FinRegLab cash-flow underwriting research (empirical and policy context) ²¹

Four sessions with detailed agendas, templates, and acceptance gates

Session-level outcomes and decision gates

Session	Primary objective	Concrete outputs by end of the 2 hours	Decision points	Acceptance criteria to proceed
Session 1	Align on business goals, AI use cases, and non-negotiable guardrails	Use-case backlog + scoring; agreed guardrails and explicit exclusions; prototype scope (1–2 priority use cases); decision log started	Select (a) top use case; (b) top 2–3 data categories to evaluate first	Guardrails signed off by Compliance/Legal Liaison; use case has defined label + success metrics; prohibited data list documented
Session 2	Decide feasible, lawful data architecture + shortlist candidate features	Data source shortlist with feasibility/latency; feature inventory v0; risk assessment matrix v0 (fair lending, privacy, explainability, third-party); data acquisition actions	Choose data sources for prototype; decide “fraud gate” vs “credit model” separation	Each shortlisted feature has (1) rationale, (2) explainability mapping, (3) risk rating, (4) documentation owner
Session 3	Design minimal technical experiments + evaluation framework	Experiment plan (dataset, joins, models, metrics); fairness-testing plan; calibration and PSI/drift plan; reason-code mapping draft; governance checkpoint plan	Approve experiment scope and sign-off for data pull; confirm evaluation thresholds	Experiment plan is reproducible; fairness metrics defined; adverse action reason feasibility confirmed for all core features
Session 4	Synthesize findings into pilot recommendation and handoff package	Pilot charter; governance and documentation pack for validation/compliance; vendor/third-party risk actions; resource estimate and timeline to pilot	Go / No-go for pilot; define success criteria and monitoring gates	Handoff package complete; validation/compliance review path defined; operational integration plan documented

Templates used across all sessions

Use these consistently to maintain auditability and align to model risk management governance expectations. ¹

Template A: Use-case scorecard (fill in Session 1)

Attribute	Notes
Use case	(e.g., originations approval; pricing; line management; early warning; collections prioritization; fraud gate)
Business KPI	(e.g., loss rate, approval rate, ROA/ROE proxy, cycle time, fraud loss)
Model outcome	(PD, EL, delinquency transition, fraud probability)
Primary decision	Approve/decline; price tier; limit tier; step-up verification; collections treatment
Explainability need	Adverse action reason codes required? (Yes/No)
Data dependency	Bureau, cash-flow, verified income, rent/utilities, etc.
Regulatory complexity	Low / Medium / High (with notes)
Implementation effort	Low / Medium / High
Priority score	1–5

Template B: Feature inventory table (fill in Session 2)

Field	Description
Feature family	Income stability / employment continuity / cash-flow resilience / expense volatility / debt burden / repayment behavior / fraud-ID
Candidate feature	Plain-language name
Data source	System/vendor and whether it may be a consumer report
Predictive rationale	Why it should predict repayment or fraud risk
Explainability mapping	Draft adverse action reason (if applicable)
Fair lending proxy risk	Low / Med / High + rationale
Privacy risk	Low / Med / High + rationale (Reg P considerations) ²²
Data quality risk	Coverage, stability, missingness, mismatch risk
Controls needed	Consent, minimization, monitoring, vendor oversight
Decision	Include / Exclude / Needs counsel review
Owner	Person responsible

Template C: Risk assessment matrix (fill in Session 2; refine Session 4)

Risk domain	Low	Medium	High	Notes / mitigations
Fair lending / proxy discrimination				
Adverse action explainability				Regulation B specificity constraint 23
Privacy / data minimization				Regulation P constraints 22
Third-party risk				Interagency lifecycle expectations 20
Security / safeguards				Safeguards program expectations 24
Data quality / stability				
Model risk / governance				SR 11-7 framework 16

Template D: Quick model evaluation checklist (fill in Session 3; confirm Session 4)

Category	Check	Evidence artifact
Label definition	☐	Label spec + observation window
Leakage controls	☐	Feature timing diagram
Baseline benchmark	☐	Baseline model metrics summary
Discrimination power	☐	AUC/KS by segment
Calibration	☐	Calibration plot + Brier score
Fairness tests	☐	Disparities table + narrative
Explainability	☐	Reason-code mapping + top drivers
Stability/drift plan	☐	PSI dashboard spec
Governance readiness	☐	Model doc outline + approvals

Session 1 agenda (2 hours): objectives, facilitation notes, decision points

Session 1 objective: Establish the AI + data opportunity space, define success metrics, and lock in guardrails (including explicit exclusions).

Agenda (120 minutes)

• 0:00-0:10 | Opening and scope check

Facilitation: Confirm product in scope (e.g., unsecured personal loans vs card vs BNPL), geography, and whether the output is a pilot recommendation vs a broader strategy backlog.

- **0:10–0:25 | “No-surprises” regulatory constraints briefing (Compliance/Legal Liaison)**

Cover: Regulation B specificity requirements; why “black box” is not a reason to avoid adverse action specificity; and the CFPB’s position that there is no AI exemption. ²⁵

- **0:25–0:45 | Lending value chain + AI opportunity mapping (Product/Ops Lead)**

Map: origination → verification → decisioning → servicing → collections. Identify where AI changes cost, cycle time, or losses (fraud, early delinquency, servicing routing).

Facilitation: Use a whiteboard with columns “Decision,” “Pain point,” “Data needed,” “AI assist (human-in-the-loop vs automated).”

- **0:45–1:05 | Use-case scoring (All; Workshop Lead moderates)**

Use Template A to score 6–10 candidate use cases and select the top 1–2 for prototype.

Decision point: Pick one **primary** use case and one **secondary** (if the primary stalls due to data constraints).

- **1:05–1:25 | Guardrails and explicit exclusions (Compliance/Legal Liaison + Lead)**

Produce the “Prohibited/High-Risk Data List” and “Permissible Data Families to Explore.” Anchor to CFPB guidance on consumer surveillance and adverse action specificity. ⁹

- **1:25–1:45 | Prototype success criteria and KPI targets (Lead + Modeler)**

Define target threshold improvements (e.g., +X AUC, improved approval at constant loss, reduced manual review time) and non-negotiables (reason codes).

Note: Interagency alternative-data guidance focuses on consumer protection implications and benefits/risks, supporting a balanced “lift + controls” framing. ¹⁹

- **1:45–2:00 | Wrap-up and assignments for Session 2**

Document decisions in a decision log and assign owners to fill Template B and Template C for the shortlisted data categories.

Outputs at end of Session 1 - Use-case backlog with scores (Template A completed) - Guardrails and explicit exclusions document (signed by Compliance/Legal Liaison) - Prototype scope charter: target outcome definition + KPIs - Decision log v1

Acceptance criteria to proceed - Clear label definition and observation window concept (even if not fully specified yet) - Explicit exclusions documented (including social graph/family/inheritance/surveillance-style signals) - Agreement on how adverse action reason codes will be produced and validated later (Reg B constraint) ²⁶

Session 2 agenda (2 hours): data architecture, feature inventory, and risk triage

Session 2 objective: Determine a lawful and feasible data architecture, shortlist candidate features, and complete a first-pass risk assessment.

Agenda (120 minutes)

- **0:00–0:10 | Reconfirm scope and guardrails**

Facilitation: Re-state exclusions to prevent “scope creep.”

- **0:10–0:30 | Data inventory review (Data Engineer)**

Present: coverage, latency, join keys, expected missingness, and vendor dependencies.

Tie-in: third-party lifecycle expectations (inventory, due diligence, ongoing monitoring) for any data/ AI vendor in scope. ²⁷

- **0:30–0:55 | Data source scoring (All)**

Score each candidate data source on: predictive promise, explainability, privacy risk, fair lending proxy risk, operational feasibility, and vendor risk.

Evidence point: interagency alternative data statement recognizes broad use cases but emphasizes consumer protection implications—this is your justification for a structured risk triage rather than “feature shopping.” ²⁸

- **0:55–1:20 | Build Feature Inventory v0 (Modeler + Data Engineer; group edits)**

Use Template B. Focus on financially grounded, interpretable feature families first (income stability, cash-flow resilience, debt burden).

Evidence point: FinRegLab’s research program documents both predictive value and policy considerations around cash-flow underwriting; this supports prioritizing cash-flow aggregates over invasive social/family signals. ²⁹

- **1:20–1:45 | Risk Assessment Matrix v0 (Compliance/Legal Liaison leads)**

Use Template C to assign Low/Med/High and mitigations.

Ensure Regulation P privacy and Safeguards/security impacts are explicitly noted for each data source category. ³⁰

- **1:45–2:00 | Decide prototype data scope + request list**

Decision point: approve the minimal dataset to pull (and what stays out until post-counsel review).

Establish governance checkpoint: “no data pull without documented purpose + minimization + access controls.”

Outputs at end of Session 2 - Data source shortlist and feasibility notes (including vendor dependencies) - Feature inventory v0 (Template B populated for ~20–40 candidate features) - Risk assessment matrix v0 (Template C populated and signed) - Prototype data request list + access plan (who, where, how logged)

Acceptance criteria to proceed - Every included feature has a draft explainability mapping (adverse action reason concept) or is explicitly marked “fraud gate only”

- High-risk features are either excluded or routed to a “needs counsel review” backlog

- Third-party risk actions exist for each vendor-dependent data source (due diligence plan) ³¹

Session 3 agenda (2 hours): minimal technical experiment plan and evaluation design

Session 3 objective: Convert business and data decisions into a minimal, auditable experiment plan with metrics (performance, calibration, fairness, drift).

Agenda (120 minutes)

- **0:00–0:10 | Confirm experiment scope and success thresholds**

Facilitation: Keep experiments minimal; avoid “science project sprawl.”

- **0:10–0:30 | Labeling and dataset schema (Modeler)**

Define: outcome labels (e.g., 60+ DPD within 12 months), observation window, censoring, and what “at decision time” means.

Tie-in: SR 11-7 stresses appropriate model development and use, which implies correct label/feature timing and avoiding misuse via leakage. ¹⁶

- **0:30–0:55 | Baseline model plan (Modeler)**

Establish baseline: logistic regression / scorecard-like model using existing bureau + application + verified income (as available).

Define incremental tests: add cash-flow aggregates, rent/utilities, etc., one bundle at a time to quantify marginal lift.

- **0:55–1:20 | Fairness, explainability, and adverse action readiness (Compliance/Legal Liaison + Modeler)**

Anchor: Regulation B requires specific principal reasons; CFPB circulars emphasize accurate reasons even for complex algorithms and warn against “check-the-box” generic reasons. ³²

Outcome: reason-code mapping draft for each feature family and a plan to validate reason accuracy.

- **1:20–1:40 | Stability/drift and monitoring design (Data Engineer + Modeler)**

Define PSI monitoring, feature drift alerts, and score distribution monitoring in pilot.

Governance tie-in: ongoing monitoring is a core expectation under SR 11-7 style frameworks and OCC model risk management examination guidance. ¹

- **1:40–2:00 | Experiment execution plan + repo/doc structure**

Decide where code lives, what constitutes an auditable run, and what artifacts are required for Session 4 decisioning.

Outputs at end of Session 3 - Minimal experiment plan (data, models, metrics, fairness tests, drift tests) - Quick model evaluation checklist (Template D ready to use) - Reason-code mapping draft (feature family → principal reasons) - Pilot monitoring spec (PSI/drift, calibration checks, fairness monitoring)

Acceptance criteria to proceed - Baseline + incremental comparisons defined (so value is measurable)

- Fairness tests defined and feasible given available demographic/proxy data governance

- Reason-code feasibility confirmed for major drivers (no reliance on vague “AI said so” explanations) ³³

Session 4 agenda (2 hours): synthesize results into pilot recommendation and handoff package

Session 4 objective: Produce the pilot recommendation, governance checkpoints, and a validation/compliance-ready handoff package.

Agenda (120 minutes)

- **0:00–0:15 | Review experiment outputs and key evidence (Modeler)**

Present: lift vs baseline, calibration, stability, subgroup metrics, reason-code sanity check.

Facilitation: If experiments were not run yet, use Session 4 to finalize the pilot plan and define the minimum evidence required before first customer impact.

- **0:15–0:35 | Risk review and mitigations (Compliance/Legal Liaison)**

Confirm: privacy constraints (Reg P), security safeguards expectations, third-party risk actions. ³⁴

- **0:35–0:55 | Governance checkpoints and documentation pack review (Workshop Lead)**

Align to SR 11-7 pillars and OCC MRM expectations for documentation, validation readiness, and change control. ¹

- **0:55–1:20 | Operational integration plan (Product/Ops Lead)**

Define rollout mechanics: channel, manual review rules, step-up verification, servicing implications, adverse action process updates, and exception handling.

Note: if using consumer reports for risk-based pricing, ensure the relevant risk-based pricing notice obligations are included in the operational plan. ³⁵

- **1:20–1:40 | Go/No-go decision and pilot success criteria (All)**

Decision point: pilot proceed, proceed with conditions, or pause pending data/controls.

• **1:40–2:00 | Finalize handoff package index + owners**

Assign owners and dates for completing remaining artifacts and reviews.

Outputs at end of Session 4 - Pilot charter (scope, KPIs, guardrails, monitoring) - Governance checkpoints schedule and sign-off matrix - Validation/compliance handoff package (index + draft contents) - Post-workshop resource estimate and pilot workplan

Acceptance criteria - Pilot recommendation includes measurable success criteria and “stop conditions”

- Documentation is sufficient for an independent validation team to reproduce results and test limitations (SR 11-7 alignment) ¹⁶

- Compliance artifacts explicitly address adverse action reason mapping and privacy/third-party controls ³⁶

Minimal technical experiment plan for prototype testing

This section consolidates (and slightly expands) the Session 3 experiment plan so it can be executed quickly and audited.

Dataset and data needs

Minimum viable dataset (MVD) - Historical applications and decisions for a defined period (e.g., 12–24 months), including application data, decision outcome, terms (APR/limit), and performance outcomes (delinquency/charge-off).

- Existing bureau/CRA attributes and internal performance variables (if already used).

- One incremental “new data source bundle” at a time (to isolate incremental value):

- Cash-flow aggregates (transaction-derived) as Bundle 1, because this is both widely discussed and empirically studied as an alternative data category. ³⁷

- Optional Bundle 2: verified income/employment data (if not already used).

- Optional Bundle 3: rent/utility/telco payment history (coverage-dependent).

Data governance constraints - Use a secure environment; do not move raw PII into unapproved tools. Privacy and security expectations under Regulation P and the Safeguards Rule should be treated as baseline constraints for the experimentation environment and access controls. ³⁰

- Log lineage (source system, refresh cadence), transformations, and access approvals to support later validation and auditability. ¹

Simple models to run (minimal but informative)

1) **Baseline:** Logistic regression (or scorecard-like binning with logistic) using incumbent data families

2) **Incremental model A:** Logistic regression + cash-flow bundle (aggregated, interpretable variables only)

3) **Incremental model B:** Gradient boosted trees (GBT) with monotonic constraints where appropriate (optional), using the same feature set as Incremental A to measure the value of nonlinearity vs data enrichment

4) **Fraud gate (if in scope):** A separate classifier for confirmed fraud/first payment default, clearly separated from credit PD modeling

The CFPB's adverse action expectations apply regardless of model complexity; complex models require the same ability to provide accurate and specific reasons. ³⁸

Evaluation metrics (including fairness, calibration, stability/drift)

Performance - AUC/ROC, KS, lift charts, approval-at-constant-loss (or loss-at-constant-approval) comparisons across models

- Business translation: expected loss impact and approval mix changes by score band

Calibration - Calibration curve, Brier score

- For pricing/limits: calibration by decision tier to ensure risk-based pricing aligns to realized performance

Fairness testing (risk management framing; not legal advice) - Outcome differences by legally permissible proxy groups or protected-class monitoring (where your governance allows), including:

- Approval rate and adverse action rate differences

- Differences in average predicted PD or score distribution

- Group-wise AUC and calibration parity tests

- Practical insight: CFPB examinations and guidance emphasize that advanced technologies do not excuse failures that lead to discrimination or opaque adverse actions; this supports treating fairness testing as a core readiness gate rather than optional analytics. ³⁹

Stability and drift - PSI for top features and model score

- Drift in missingness/coverage (common risk for alternative data and vendor feeds)

- Monitoring thresholds pre-defined for pilot rollback, consistent with ongoing monitoring expectations in model risk management frameworks. ¹

Prohibited/high-risk data and feature flags (for the experiment)

Use this table to keep the modeler and data engineer aligned on what **not** to test as underwriting drivers.

Category	Examples	Why to avoid / treat as presumptively impermissible for creditworthiness
Social relationships & interactions	Social graph, friend network risk, engagement/likes, message metadata	High proxy discrimination risk; invasive; hard to explain with specific principal reasons; CFPB warns about use of consumer surveillance data and stresses no AI exemption for specificity ⁹
Family connections	Household network, relatives' repayment histories, inferred kinship	Often acts as a proxy for protected characteristics/ socioeconomic status; weak business necessity; adverse action explanation challenges
Inheritance / expected wealth	"Inheritance potential," expected transfers	Speculative, inequitable, not a repayment-capacity measure; difficult to defend and explain

Category	Examples	Why to avoid / treat as presumptively impermissible for creditworthiness
Consumer surveillance / sensitive behavioral trails	Continuous location trails, app usage unrelated to finances	CFPB notes underwriting may use surveillance-harvested data and that consumers may be denied for reasons not intuitively related; increases privacy and explainability risk ³⁹
Device signals as credit risk	Device fingerprint as PD driver	Proxy-prone; better suited to fraud/ID gating with step-up verification rather than PD

Governance checkpoints, documentation requirements, and validation/compliance handoff

Governance checkpoints (recommended gates)

These checkpoints are designed to be consistent with interagency model risk management expectations and interagency third-party risk management standards:

1) Pre-data-pull gate (after Session 2)

- Document purpose, minimization, and access controls (privacy/security baseline). ⁴⁰
- Third-party due diligence plan for vendors/data aggregators in scope. ²⁰

2) Pre-model-run gate (after Session 3)

- Label/feature timing review to prevent leakage (model risk). ¹
- Confirm adverse action reason-code feasibility for all major feature families (Reg B specificity). ³³

3) Pre-pilot gate (Session 4 output)

- Evidence pack includes performance, calibration, fairness analyses, stability/drift readiness, and limitations; supports independent validation scope. ¹⁶
- Operational readiness: adverse action workflows, risk-based pricing notices if applicable, and monitoring/rollback triggers. ³⁵

Documentation requirements (what must exist before a pilot)

A “pilot-ready” package should typically include:

- **Model development document:** purpose, scope, data sources, feature definitions, training approach, limitations, and monitoring plan (SR 11-7 alignment). ¹
- **Explainability and adverse action mapping:** feature-family-to-reason-code taxonomy; process to ensure reasons “relate to and accurately describe” the factors considered; guidance to keep reasons concise. ⁴¹
- **Fair lending testing memo:** the tests performed, results, mitigations, and residual risks. (The interagency fair lending exam procedures provide examiner frameworks and reinforce a structured approach to fair lending risk management.) ⁴²

- **Privacy and security artifacts:** data handling narrative aligned to Regulation P; information security controls aligned to Safeguards expectations. ⁴³
- **Third-party risk file:** vendor inventory entry, due diligence, contracting controls, ongoing monitoring plan. ²⁰

Handoff package index (deliver this to independent validation/compliance)

- 1) Decision log (all major choices and rationales)
- 2) Data inventory and lineage diagrams
- 3) Feature inventory + risk assessment matrix
- 4) Experiment plan + reproducible run instructions
- 5) Results pack (metrics, calibration, fairness, drift readiness)
- 6) Reason-code mapping and adverse action readiness memo
- 7) Monitoring plan + thresholds + rollback plan
- 8) Vendor/third-party risk documentation (if applicable)
- 9) Pilot charter and operational integration plan

Estimated effort and resource checklist to reach a pilot after the workshop

This section is intentionally practical: it assumes the workshop produces a “pilot recommendation + plan,” and you need to estimate the build.

Typical post-workshop workstreams and effort ranges

Effort varies widely by whether you already have cash-flow access, verification vendors, and a mature model governance program. The ranges below assume modest existing infrastructure but not a fully built alternative-data pipeline.

Workstream	Key tasks	Typical effort (person-hours)	Primary owners
Data contracting & third-party onboarding	Vendor due diligence, contracting, security assessment, monitoring plan	40–120	Compliance/Legal Liaison + Data Engineer
Data engineering & feature pipeline	Data ingestion, quality checks, lineage logging, feature computation, refresh cadence	80–200	Data Engineer
Modeling & experimentation	Baseline + incremental models, tuning, calibration, documentation of limitations	60–160	Modeler
Fair lending / explainability work	Proxy screening, disparity analyses, reason-code validation, mitigation iterations	40–120	Compliance/Legal Liaison + Modeler

Workstream	Key tasks	Typical effort (person-hours)	Primary owners
Governance & validation readiness	Model documentation, inventory entry, validation coordination, change control plan	30-100	Workshop Lead
Ops integration	Decision workflow updates, verification steps, adverse action process updates, monitoring dashboards	40-120	Product/Ops Lead

The emphasis on documenting development, validation, governance, and controls reflects interagency model risk management expectations. ¹

Third-party lifecycle controls align with interagency third-party guidance. ²⁰

Resource checklist (what you need in place to pilot safely)

- **Secure analytics environment** with controlled access, audit logging, and clear rules on use of AI tools (no raw PII in unapproved systems), aligned to privacy/security expectations. ⁴³
- **Data lineage and feature dictionary** sufficient for independent validation reproduction. ¹
- **Adverse action reason-code library** mapped to feature families; process to ensure reasons are specific and reflect actual drivers. ⁴⁴
- **Monitoring dashboards** for performance, calibration, drift (PSI), and fairness indicators, plus rollback triggers. ¹
- **Vendor oversight artifacts** for any new data/AI provider: due diligence, contracts, ongoing monitoring. ²⁰
- **Pilot governance:** defined approval authority, exception policy, and stop conditions consistent with model risk management governance expectations. ¹

Practical “pilot readiness” definition (what “ready” means)

A pilot should be considered ready when: - The team can demonstrate incremental value over baseline (or a credible hypothesis with a clear test plan if live data is required)

- Adverse action explanations are demonstrably specific and accurate for anticipated actions (no generic “bucket” reasons) ⁴⁵

- Fair lending and privacy risks are documented with mitigations and governance owners

- Independent validation has enough artifacts to reproduce results and challenge assumptions, consistent with SR 11-7 style expectations ¹⁶

- Third-party risk governance is in place for any external data/AI components ³¹

¹ ² ¹⁶ Supervisory Guidance on Model Risk Management

https://www.federalreserve.gov/supervisionreg/srletters/sr1107a1.pdf?utm_source=chatgpt.com

³ ¹⁵ ²³ ²⁵ ²⁶ ³² ³³ ⁴⁴ 12 CFR 1002.9 -- Notifications.

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